

Palette-PrintAR: an augmented reality fluidic design tool for multicolor resin 3D printing

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1 INTRODUCTION

While 3D printing affords designers unprecedented geometrical complexity, fewer design tools for multi-material platforms exist, and those that have been proposed for resin printing lack interactivity. Here, we report our ongoing work leveraging our novel multi-material 3D printing method [7], which prints channels concurrently with the object itself, to empower designers to, with minimal hardware changes, paint 3D objects in real time. Our framework allows users to print a fluidic network to distribute different colors spatiotemporally into the vat and control this fabrication process with an augmented reality (AR) design and simulation tool, which we term *Palette-PrintAR*. Below, we describe the mode of user interaction with our tool, describe our underlying simulation approach, and report initial experimental validation.

2 RELATED WORK

2.1 Augmented reality and fabrication

The combination of AR with personal fabrication has emerged as a topic of interest among HCI researchers, with pioneering work such as InfraredTags, which embeds AR tags in printed objects [2]. Others have proposed augmenting printed objects after fabrication [11] and integrating real-time control in AR [13]. Jeong et al. introduce "Mechanism Perfboard", supporting linkage design for physical computing [5]. Weichel et al. in "MixFab" seek to democratize 3D printing by substituting 3D modeling with design in mixed reality [22]. "Printy3D" allows users to design modular structures using AR feedback [24], Xie et al. use spatial AR to help users design elastic structures for fabrication [23], and Hattab et al. develop a spatial AR tool to aid 3D model carving [4].

2.2 Multimaterial and multicolor 3D printing

The expanded possibilities for interactive products with multimaterial 3D printing has motivated HCI researchers to develop novel

design tools for such technologies. These include "Multi-ttch" for compositive attachments [6], "Foundry" for hierarchical material design [21], "Colormod" for digitally painting objects using photochromic inks [15], and "Pop-up print" for multimaterial distributions in shape-changing products [12]. Others combine 3D printing and spray coating [3], fabricate viewpoint-dependent lenticular surfaces [25], derive methods for printing multimaterial filaments [19], and propose automated filament inking for multi-color extrusion [8]. While multimaterial resin printers have been proposed [1, 17], these lack interactivity. Applications include objects made interactive with capacitive touch sensing [16], shape-changing "shrink-cells" [10], and customizable wearable products and fashion [18].

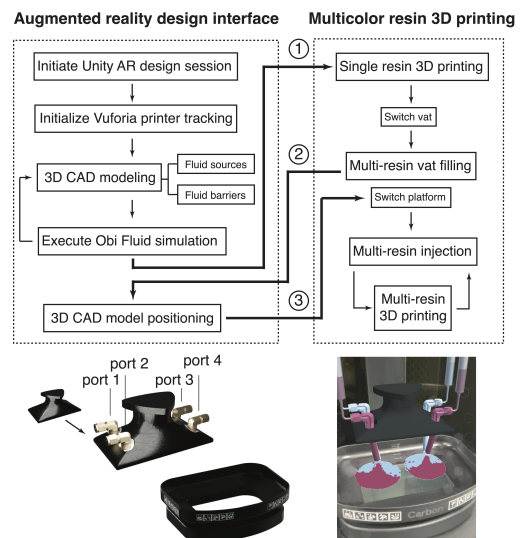


Figure 1: State diagram describing user interaction with AR fluidic design tool and multicolor resin printing.

3 AR-ENABLED MULTI-COLOR PRINTING

Our system architecture for AR-enabled multi-color printing is summarized in **Figure 1**. Interactions between the AR experience and fabrication are noted as (1) printing of AR-designed fluidic network; (2) injection of resins through AR-designed channels; and, (3) multicolor printing of AR-oriented objects. At runtime, at the beginning of a design session, the user aligns a virtual overlay of the outline of the printer component, but can then move and pan around the printer.

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3.1 Stage 1: Fluidic channel design in AR

Unlike any other printing, injection 3d printing method allows designers to specify any nozzle geometry by printing the fluidic network itself. After printer tracking initiates, the user begins by designing a series of channels to distribute multiple different resins for subsequent printing (**Figure 2**). They do so with a real model of the vat, which serves as a palette, and a virtual model of the injection platform. If user wishes to maintain sharper boundaries between injected resin, and hence in multi-color printed parts, wall structures in addition to distributor network channels can be prototyped in the AR design session.

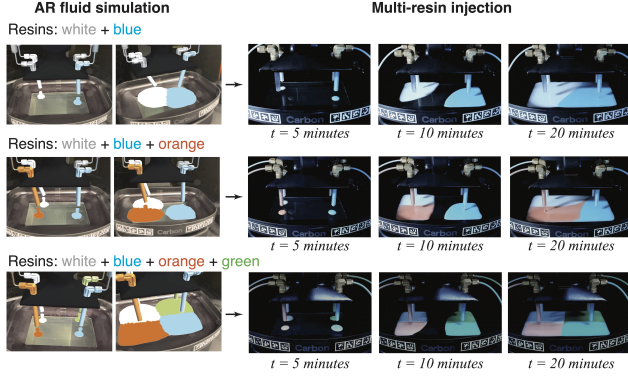


Figure 2: For two (top), three (middle) and four (bottom) resins, the user prototypes fluid flow in AR.

3.2 Stage 2: Fluid simulation in AR

The user can then prototype different distributions of resin before printing, using Multiphysics Obi fluid simulation as back-end for on-device real-time performance. Subsequently, using the single material printed distributor network, multiple resins are injected into the vat spatiotemporally. The user can also mix colors by design, as shown in **Figure 3**. Here, the user positions injection channels proximate to each other in AR design session. Subsequently, during multi-resin injection, two resins are injected. After positioning parts to be printed in AR, multicolor fabrication resumes.

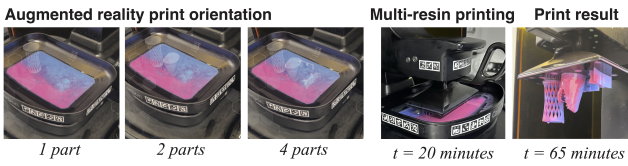


Figure 3: Resin color mixing by design.

3.3 Stage 3: Object placement in AR

Given it is challenging to predict precisely fluid flows in the vat, we incorporate into our AR tool a module that allows users to position and orient objects after vat filling, but before printing begins. An example such interaction is shown in **Figure 4**. After injection of multiple resins, user positions parts to be printed in the vat using

the augmented reality interface. A new platform is then lowered and printing begins. Illustrative multicolor print results by our method are shown in **Figure 5**. After printing, the user can simply mix the injected resins and reuse, meaning the process is waste-free.

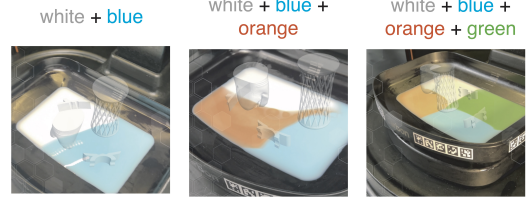


Figure 4: Orienting objects in AR before printing.



Figure 5: Illustrative multicolor print results by our method.

3.4 Materials and Methods

3.4.1 Hardware add-on. For our printer, we utilize a modified M1 printer from Carbon3D, with one hardware modification, though we emphasize our approach can be applied to any stereolithography (SLA)-style printer. Specifically, we hack this commercially available printer (build area 540 x 654 x 1734 mm) by drilling four holes in the platform. We use a standard prototyping resin (UMA 90), sold in varying hues or with dyed additives, for all our experiments.

3.4.2 Augmented reality software. We leverage Unity Game Engine (version 2021.3.25f) to develop our tool [20], which we include in the Supplement. For fluid simulation, we use Obi Fluid Physics, a CPU-based realtime particle physics engine for Unity [9]. To realistically overlay design data in the physical printer environment, we develop our tracking module with Vuforia (version 10.15.4) developed by PTC Inc., an augmented reality (AR) software development kit (SDK) [14]. Specifically, we track the pose of the printer vat, whose CAD model is available and we use to train an AI model to recognize the object position and orientation during 360 degree tracking; we endow the vat with Aruco markers to enhance tracking.

4 FUTURE DIRECTIONS

Future directions for this work will focus on leveraging novel augmented and mixed reality devices to enhance the user experience with our multicolor printing system, closing the loop between AR design decisions such as channel design and flow rate with hardware control, and exploring interactive multimaterial devices that can be fabricated by this approach.

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